US 20250277828A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0277828 A1
(43) Pub. Date: Sep. 4, 2025
BASU et al.

(54) **CURRENT SENSOR CIRCUIT WITH DIFFERENCE AMPLIFIER**

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(21) Appl. No.: **18/591,877**

(22) Filed: **Feb. 29, 2024**

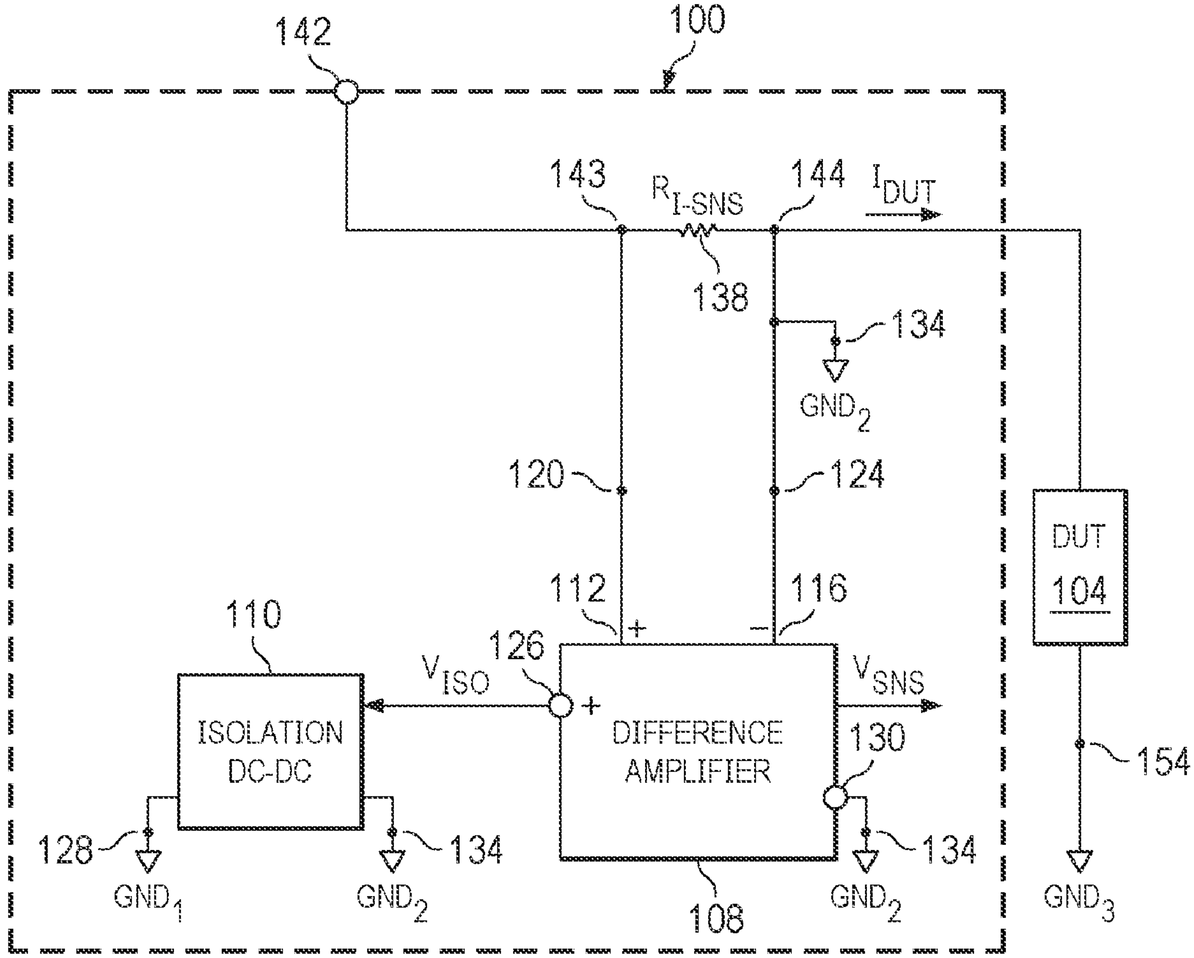
Publication Classification

(51) **Int. Cl.**
G01R 19/25 (2006.01)
H03F 3/45 (2006.01)

(52) **U.S. Cl.**
CPC ***G01R 19/25*** (2013.01); ***H03F 3/45475*** (2013.01)

(57) **ABSTRACT**

A current sensor circuit includes a sensing resistor having a first node adapted to be coupled to a high side terminal of a voltage supply and a second node adapted to be coupled to a high side terminal of a DUT (device under test). A low side terminal of the DUT is coupled to a first ground node. The current sensor circuit includes a difference amplifier coupled to the first node and the second node of the sensing resistor. A low side power terminal of the difference amplifier is coupled to a second ground node, and the second ground node is coupled to the high side terminal of the DUT. The current sensor circuit also includes an isolation DC (direct current)-to-DC converter having an output coupled to a high side power terminal of the difference amplifier.



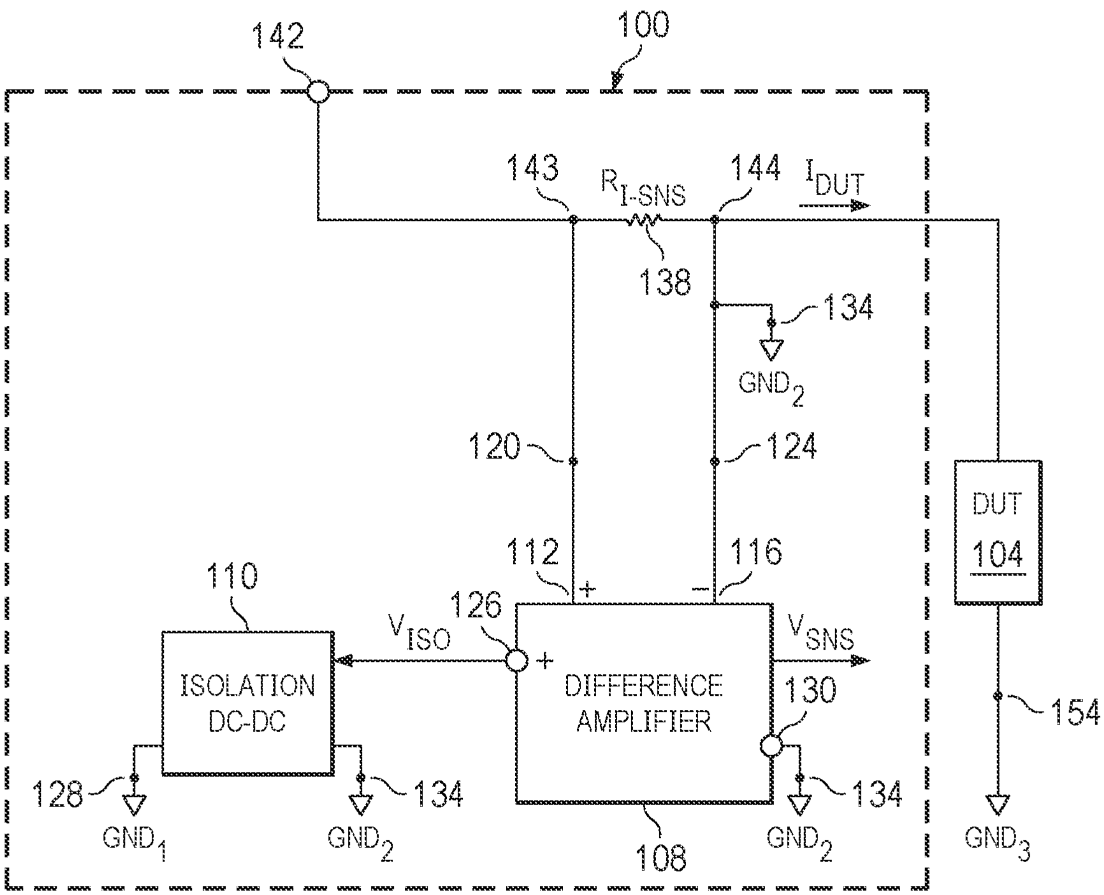

142
100
143
$R_{I\text{-}SNS}$
144
I_{DUT}
138
134
GND_2
120
124
DUT
104
112
116
110
126
V_{ISO}
+
−
+
V_{SNS}
ISOLATION
DC-DC
DIFFERENCE
AMPLIFIER
130
154
128
134
134
GND_1
GND_2
108
GND_2
GND_3

FIG. 1

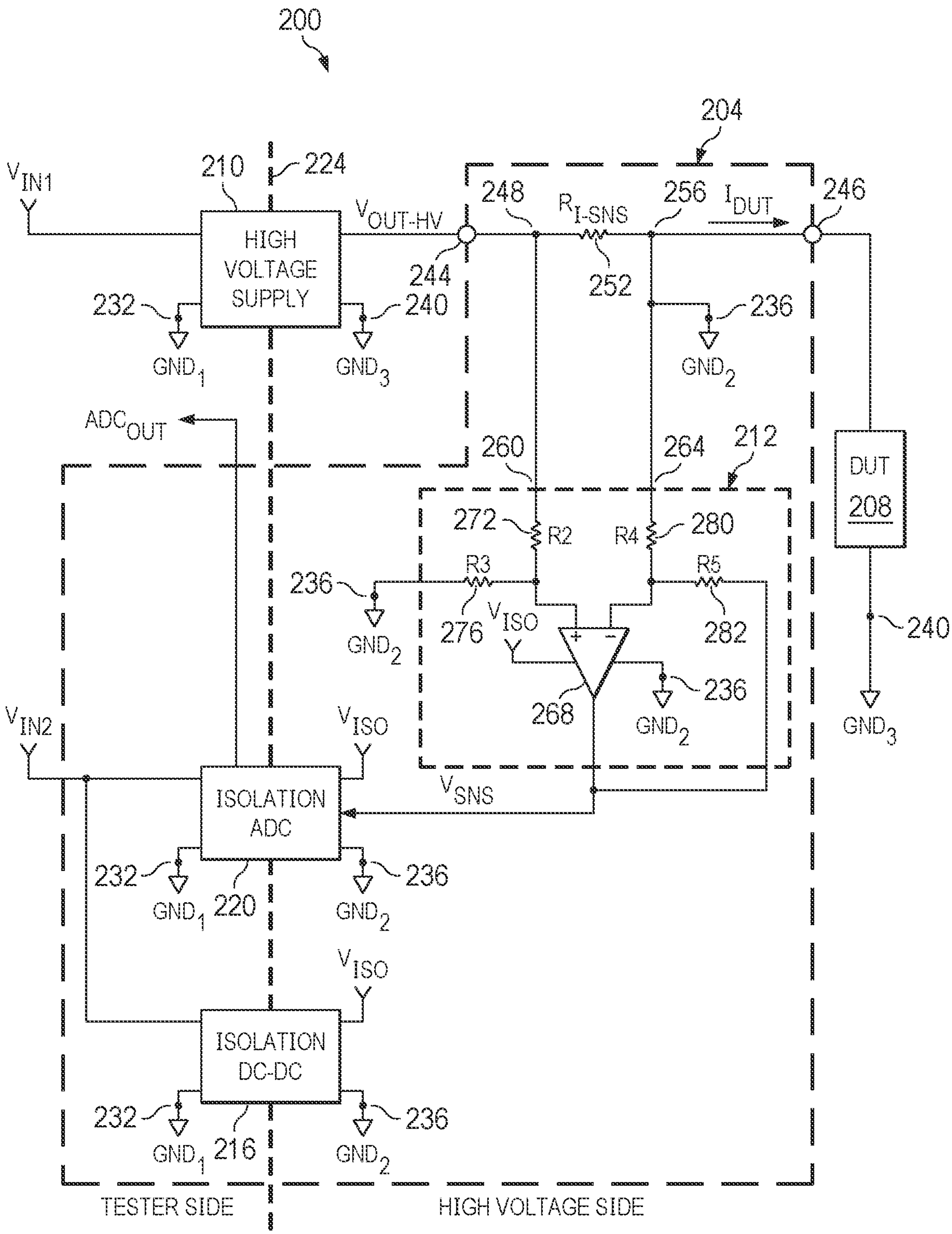

200
204
210
224
V_IN1
HIGH VOLTAGE SUPPLY
V_OUT-HV
248
R_I-SNS
256
I_DUT
246
244
252
232
240
236
GND_1
GND_3
GND_2
ADC_OUT
260
264
212
DUT
208
272
R2
R4
280
236
R3
R5
V_ISO
276
282
240
GND_2
268
236
GND_2
GND_3
V_IN2
V_ISO
ISOLATION ADC
V_SNS
232
236
GND_1
220
GND_2
V_ISO
ISOLATION DC-DC
232
236
GND_1
216
GND_2
TESTER SIDE
HIGH VOLTAGE SIDE

FIG. 2

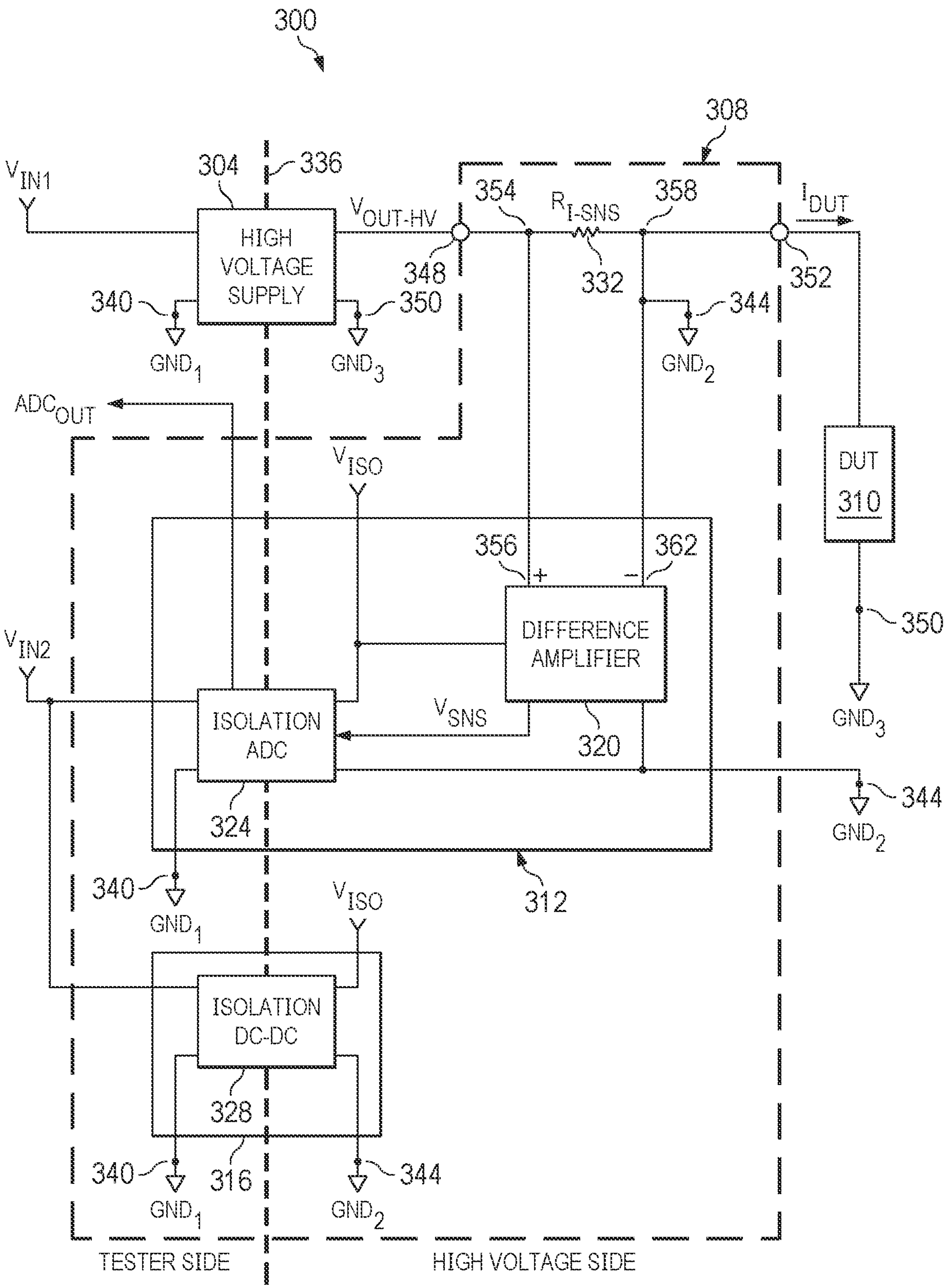

300
308
V_{IN1}
304
336
HIGH VOLTAGE SUPPLY
V_{OUT-HV}
354
R_{I-SNS}
358
I_{DUT}
348
332
352
340
350
344
GND_1
GND_3
GND_2
ADC_{OUT}
V_{ISO}
DUT
310
356
362
+
−
DIFFERENCE AMPLIFIER
350
V_{IN2}
ISOLATION ADC
V_{SNS}
320
GND_3
324
344
GND_2
340
GND_1
312
V_{ISO}
ISOLATION DC-DC
328
340
316
344
GND_1
GND_2
TESTER SIDE
HIGH VOLTAGE SIDE

FIG. 3

400

VHV/VCM -GND3(V)	AMMETER CURRENT FLOWING FROM VHV TO GND-3(uA)	MEASURED CURRENT WITH CURRENT SENSING CIRCUIT (uA)	CURRENT MEASURE ERROR (uA)
105.0	5.201	5.759375	-0.558375
201.4	9.97	10.85625	-0.88625
396.9	19.65	20.296875	-0.646875
788.1	39.05	39.4328125	-0.3828125
984.9	48.76	49.340625	-0.580625
1478.4	73.19	73.874054	-0.684054

FIG. 4

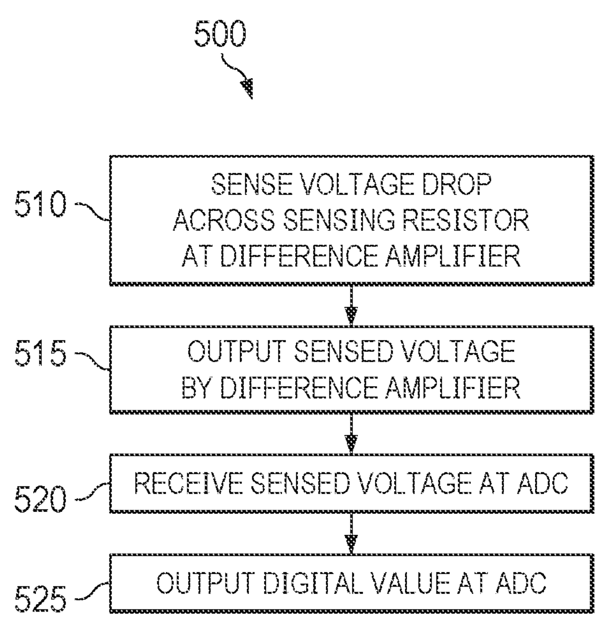


FIG. 5

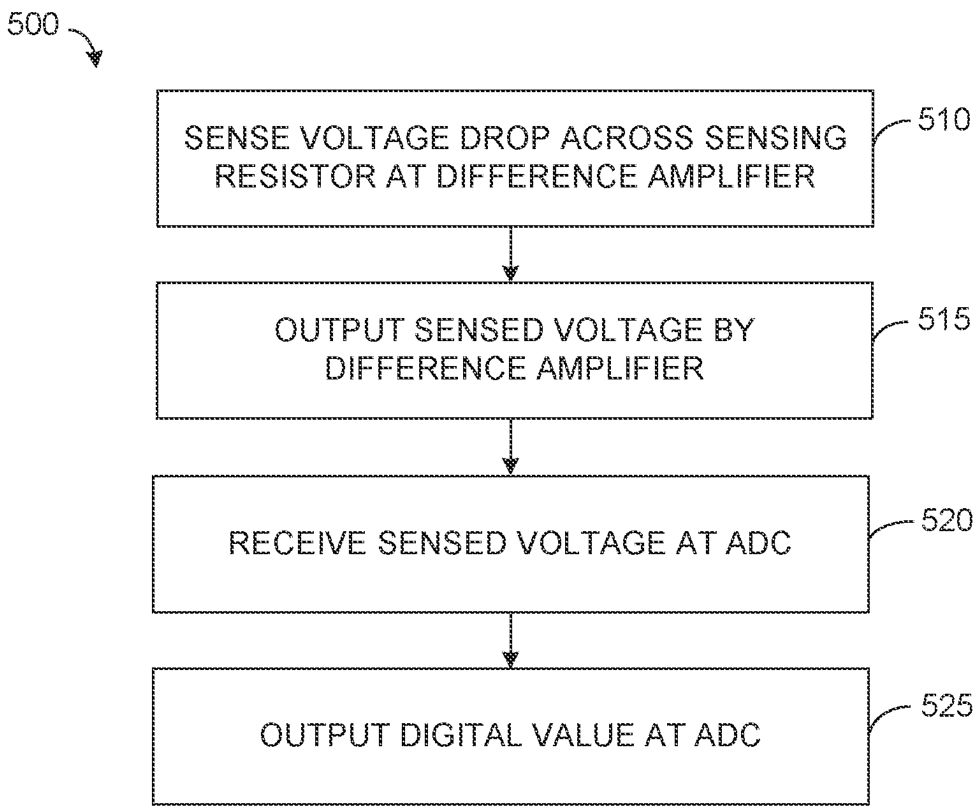

500
SENSE VOLTAGE DROP ACROSS SENSING RESISTOR AT DIFFERENCE AMPLIFIER
510
OUTPUT SENSED VOLTAGE BY DIFFERENCE AMPLIFIER
515
RECEIVE SENSED VOLTAGE AT ADC
520
OUTPUT DIGITAL VALUE AT ADC
525

FIG. 5

CURRENT SENSOR CIRCUIT WITH DIFFERENCE AMPLIFIER

TECHNICAL FIELD

[0001] This description relates to a circuit for sensing current that includes a difference amplifier.

BACKGROUND

[0002] A difference amplifier, also known as a differential amplifier, is an electronic device that amplifies the difference between two input voltages while rejecting voltage that is common to both inputs (common mode voltage). Difference amplifiers are employed in situations where a signal of interest is the voltage difference between two points, which may be superimposed on the common-mode voltage that is not of interest and should be curtailed.

[0003] An operational amplifier, commonly known as an op-amp, is a type of voltage amplifier that is designed to amplify the difference between two input voltage levels. Op-amps typically has a very high gain and are used in a variety of configurations depending on the application.

[0004] Placing a sensing resistor in series with a circuit under test offers a straightforward method for measuring current, leveraging the fundamental principle of Ohm's Law. In this setup, the sensing resistor is inserted directly into a current path of the circuit under test. As current flows through this sensing resistor, a voltage drop across the sensing resistor is directly proportional to the amount of current across the resistor and to the current provided to the circuit under test due to the relationship defined by Ohm's Law (V=IR, where V is the voltage across the resistor, I is the current through the resistor, and R is the resistance). By measuring this voltage drop and knowing the resistance of the sensing resistor, the current flowing through the circuit can be accurately calculated.

SUMMARY

[0005] A first example relates to a current sensor circuit that includes a sensing resistor having a first node adapted to be coupled to a high side terminal of a voltage supply and a second node adapted to be coupled to a high side terminal of a DUT (device under test). A low side terminal of the DUT is coupled to a first ground node. The current sensor circuit includes a difference amplifier coupled to the first node and the second node of the sensing resistor. A low side power terminal of the difference amplifier is coupled to a second ground node, and the second ground node is coupled to the high side terminal of the DUT. The current sensor circuit also includes an isolation DC (direct current)-to-DC converter having an output coupled to a high side power terminal of the difference amplifier.

[0006] A second example relates to a current sensing circuit that includes a sensing resistor having a first node adapted to be coupled to a high side terminal of a voltage supply and a second node adapted to be coupled to a high side terminal of a DUT. A low side terminal of the DUT is coupled to a first ground node. The current sensing circuit includes a difference amplifier coupled to the first node and the second node of the sensing resistor. A low side power terminal of the difference amplifier is coupled to a second ground node that is galvanically isolated from the first ground node, and the second ground node is coupled to the high side terminal of the DUT. The difference amplifier is configured to output a sensed voltage that varies as a function of a current across the sensing resistor. The current sensing circuit also includes an isolation DC-to-DC converter configured to provide an isolated voltage to a high side power terminal of the difference amplifier.

[0007] A third example relates to a method for sensing current. The method includes sensing, by a difference amplifier, a voltage drop across a sensing resistor having a first node adapted to be coupled to a high side terminal of a voltage supply and a second node adapted to be coupled to a high side terminal of a DUT. A low side terminal of the DUT is coupled to a first ground node, the difference amplifier is coupled to the first node and the second node of the sensing resistor and a low side power terminal of the difference amplifier is coupled to a second ground node. The second ground node is coupled to the high side terminal of the DUT. The method also includes outputting, by the difference amplifier, a sensed voltage that varies as a function of a current across the sensing resistor in response to receiving an isolated voltage at a high side power terminal of the difference amplifier provided from an isolation DC (direct current)-to-DC converter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** illustrates an example of a current sensor circuit for sensing current provided to a DUT (device under test).

[0009] FIG. **2** illustrates an example of a circuit that includes a current sensor circuit that measures a current provided to a DUT based on a voltage provided by a high voltage supply.

[0010] FIG. **3** illustrates an IC (integrated circuit) package level circuit that is employable to implement the circuit of FIG. **2**.

[0011] FIG. **4** illustrates a chart with measured results from implemented circuits.

[0012] FIG. **5** is a flowchart of an example method of sensing a current.

DETAILED DESCRIPTION

[0013] This description relates to a circuit for measuring current that measures a voltage drop across a sensing resistor connected to a DUT (device under test). Conventional methods for measuring current encounter significant challenges when the common mode voltage between resistor terminals is high, particularly in applications exceeding 400 volts (V). Standard circuit components, such as transistors and op-amps (operational amplifiers) are not designed to handle such high common mode voltages.

[0014] This description provides a current sensor circuit capable of accurately measuring current for nearly any voltage, including high voltage applications, potentially up to and exceeding 2000 volts. The current sensor circuit employs an isolated power supply for a difference amplifier implemented with an op-amp, which allows the op-amp to operate without exposure to high common mode voltages. The current sensor circuit has separate ground references for the DUT and for the difference amplifier, effectively isolating the difference amplifier component from a high common mode voltage.

[0015] The current sensor circuit includes a sensing resistor, the difference amplifier, and an isolation DC (direct current)-to-DC converter. The sensing resistor is in series with the DUT (device under test), and the difference amplifier is configured to measure the voltage drop across this resistor. The isolation DC-to-DC converter provides an isolated voltage to a high side power terminal of the difference amplifier, ensuring that the difference amplifier operates within common mode voltage limits.

[0016] Additionally, in some examples, the current sensor circuit includes an isolation ADC (analog-to-digital converter) to digitize the sensed voltage, allowing for digital processing and readout. In some examples, the isolation ADC and the difference amplifier are implemented on a first IC (integrated circuit) package, and the isolation DC-to-DC converter can be implemented on a second IC package, further enhancing isolation between the high voltage and low voltage sides of the circuit. The current sensor circuit measures current at high voltages without the limitations of conventional approaches making this approach suitable for applications such as, high voltage electric vehicles and other high voltage systems where accurate current measurement is needed.

[0017] FIG. **1** illustrates an example of a current sensor circuit **100** for sensing current provided to a DUT **104**. The DUT **104** represents a load that has an on-state and an off-state. For simplification, it is presumed that in the on-state, a DUT current, I_{DUT} provided to the DUT **104** is at a maximum level. Conversely, in an off-state, the DUT current, I_{DUT} provided to the DUT **104** defines a leakage current, I_{LEAK} for the DUT **104**. In general, the leakage current, should be curtailed. However, for high voltage applications, it can be difficult to accurately measure the leakage current, I_{LEAK} for the DUT **104** due to a high common mode voltage, V_{CM} of conventional current measurement devices. However, the current sensor circuit **100** is configured to ensure that the common mode voltage, V_{CM} is about 5 volts (V) or less, such that standard, and precise circuit components are employable to measure the DUT current, I_{DUT} provided to the DUT **104** in both the on-state and the off-state.

[0018] The current sensor circuit **100** includes a difference amplifier **108** and an isolation DC-to-DC converter **110** (labeled ISO DC-DC). In some examples, the difference amplifier **108** is implemented with an op-amp configured as a differential amplifier for difference amplification. The difference amplifier **108** includes a non-inverting input **112** (e.g., a first input) and an inverting input **116** (e.g., a second input). The non-inverting input **112** is coupled to a first node **120** and the inverting input **116** is coupled to a second node **124**. A high side power terminal **126** of the difference amplifier **108** is coupled to an output of the isolation DC-to-DC converter **110**. The isolation DC-to-DC converter **110** is also coupled to a first ground node **128**, GND_1. A low side power terminal **130** of the difference amplifier **108** is coupled to a second ground node **134**, GND_2. The inverting input **116** of the difference amplifier **108** and the isolation DC-to-DC converter **110** are also coupled to the second ground node **134**. The first ground node **128** and the second ground node **134** are galvanically isolated from each other.

[0019] The current sensor circuit **100** includes a sensing resistor **138**, $R_{I\text{-}SNS}$. The sensing resistor **138** has a relatively small resistance, such as about 10 milliohms (mΩ) or less. The sensing resistor **138** is coupled to the first node **120** and the second node **124**. Additionally, a high side terminal of the DUT **104** is coupled to the second node **124**. The second node **124** is adapted to be coupled to a high side terminal **142** (e.g., an output terminal) of a voltage supply (not shown), such that the sensing resistor **138** is adapted to be coupled to the high side terminal **142** of the voltage supply. More particularly, the sensing resistor **138** has a first node **143** coupled to the first node **120**, and the first node **143** is adapted to be coupled to the high side terminal **142**. The sensing resistor **138** also has a second node **144** coupled to the second node **124**, and the second node **144** is adapted to be coupled to the high side terminal of the DUT **104**. In this manner, due to the high input impedance of the non-inverting input **112** and the inverting input **116** of the difference amplifier **108**, the sensing resistor **138** is in series with the DUT **104**.

[0020] A low side terminal of the DUT **104** is coupled to a third ground node **154**, GND_3. The third ground node **154** is galvanically isolated from the second ground node **134** and the first ground node **128**. Moreover, as illustrated, the high side terminal of the DUT **104** is coupled to the second ground node **134** (as well as the second node **124**).

[0021] In operation, the voltage supply outputs a voltage on the high side terminal **142**. Additionally, the isolation DC-to-DC converter **110** outputs an isolation voltage, V_{SIO} to the high side power terminal **126** of the difference amplifier **108**. The voltage provided at the high side terminal **142** can be nearly any voltage. In particular, the voltage at the high side terminal **142** is a DC voltage and can be a high voltage of about 400 volts (V) to about 2000 V (or even exceeding 2000 V in some examples). The voltage at the high side terminal **142** induces a DUT current, I_{DUT}. The difference amplifier **108** is configured to measure a voltage drop across the sensing resistor **138** and output a sensing voltage, V_{SNS} that varies as a function of a current across the sensing resistor **138**, which corresponds to the DUT current, I_{DUT}.

[0022] As noted, the DUT current, I_{DUT} flows to the third ground node **154**. Additionally, because the inverting input **116** is coupled to the second ground node **134**, and the second ground node **134** and the third ground node **154** are separate grounds (galvanically isolated), the common mode voltage between the non-inverting input **112** and the inverting input **116** is about 5 V or less with respect to the difference amplifier **108** and the second ground node **134**, such that standard components, such as an op-amp is employable to implement the difference amplifier **108**.

[0023] By use of the current sensor circuit **100**, the DUT current, I_{DUT} is accurately measured, particularly for voltages of 400 V or more without requiring high voltage components for the difference amplifier **108**. Moreover, the current sensor circuit **100** measures both on and off-state leakage current provided to the DUT **104**.

[0024] FIG. **2** illustrates an example of a circuit **200** that includes a current sensor circuit **204** that measures a current provided to a DUT **208** based on a voltage provided by a high voltage supply **210**. The DUT **208** can be representative of a load, such as a test circuit or test IC package that is driven by the high voltage supply **210**. The DUT **208** has an on-state and an off-state. For simplification, it is presumed that in the on-state, a DUT current, I_{DUT} provided to the DUT **208** is at a maximum level. Conversely, in an off-state, the DUT current, I_{DUT} provided to the DUT **208** defines a leakage current, I_{LEAK} for the DUT **208**. In general, the leakage current, should be curtailed. However, for high voltage applications, it can be difficult to accurately measure the leakage current, I_{LEAK} for the DUT **208** due to a high common mode voltage, V_{CM} of conventional current measurement devices. However, the current sensor circuit **204** is configured to ensure that the common mode voltage, V_{CM} is about 5 volts (V) or less with respect to components of the current sensor circuit **204**, such that standard, and precise circuit components are employable to measure the DUT current, I_{DUT} provided to the DUT **208** in both the on-state and the off-state.

[0025] The current sensor circuit **204** is employable to implement the current sensor circuit **100** of FIG. **1**. The current sensor circuit **204** includes a difference amplifier **212** that is employable to implement the difference amplifier **108** of FIG. **1**. The current sensor circuit **204** also includes an isolation DC-to-DC converter **216** (labeled ISO DC-DC) that is employable to implement the isolation DC-to-DC converter **110** of FIG. **1**. The current sensor circuit **204** also includes an isolation ADC **220** (analog to digital converter, labeled ISO ADC).

[0026] The current sensor circuit **100** includes a dashed line **224** to denote a tester side and a high voltage side. In the example illustrated, voltages input and output on the tester side are less than about 20 V. Additionally, the voltages input and output on the high voltage side can be any voltage, including high voltages of at least about 400 V or greater (including 2000 V or more). Some components, such as the isolation DC-to-DC converter **216**, the isolation ADC **220** and the high voltage supply **210** provide an interface between the tester side and the high voltage side.

[0027] The isolation DC-to-DC converter **216** is coupled to a first ground node **232**, GND_1 and a second ground node **236**, GND_2. In some examples, the high voltage supply **210** is coupled to the first ground node **232** and a third ground node **240** GND_3. In other examples, the high voltage supply **210** is not coupled to the first ground node **232** and is coupled to the third ground node **240** (e.g., the high voltage supply **210** is not an isolated voltage supply). The first ground node **232** and the second ground node **236** and the third ground node **240** are galvanically isolated from each other.

[0028] The high voltage supply **210** is coupled to a first input voltage, V_{IN1} provided from a DC supply (not shown). In some examples, the first input voltage, V_{IN1} is about 12 V or less. The isolation DC-to-DC converter **216** and the isolation ADC **220** are coupled to a second input voltage, V_{IN2} provided from a DC supply (not shown). The second input voltage, V_{IN2} is about 3 V to about 12 (e.g., about 5 V, in some examples). The isolation DC-to-DC converter **216** outputs an isolation voltage, V_{ISO} that is about 5 V above the second ground node **236**, such that the isolation voltage, V_{ISO} is output with respect to the second ground node **236**. The isolation voltage, V_{ISO} is output to the difference amplifier **212** and the isolation ADC **220**.

[0029] The high voltage supply **210** outputs a high voltage output (V_{OUT-HV}) to a high side terminal **244** of the current sensor circuit **204**. The high side terminal **244** of the current sensor circuit **204** is coupled to a first node **248** of the current sensor circuit **204**. The first node **248** of the current sensor circuit **204** is coupled to a sensing resistor **252**, R_{I-SNS}. The sensing resistor **252** has a resistance of about 10 mΩ or less. The sensing resistor **252** is also coupled to a second node **256** of the current sensor circuit **204**. The second node **256** is coupled to the second ground node **236** through a low side terminal **246** of the current sensor circuit **204** and to a high side terminal of the DUT **208**. A low side terminal of the DUT **208** is coupled to the third ground node **240**.

[0030] The difference amplifier **212** has a first input **260** (e.g., a non-inverting input) coupled to the first node **248** and a second input **264** (e.g., an inverting input) coupled to the second node **256**. The first input **260** is employable to implement the non-inverting input **112** of the difference amplifier **108** of FIG. **1**. Similarly, the second input **264** is employable to implement the inverting input **116** of the difference amplifier **108** of FIG. **1**. The difference amplifier **212** is implemented with an op-amp **268** arranged in a differential configuration. A high side power terminal of the op-amp **268** receives the isolation voltage, V_{ISO} from the isolation DC-to-DC converter **216**. A low side power terminal of the op-amp **268** is coupled to the second ground node **236**. Due to the high input impedance of the non-inverting input and the inverting input of the op-amp **268**, the sensing resistor **252** is in series with the DUT **208**.

[0031] The first input **260** of the difference amplifier **212** is coupled to a second resistor **272**, R**2**. The second resistor **272** is also coupled to a non-inverting input of the op-amp **268**. The non-inverting input of the op-amp **268** is also coupled to a third resistor **276**, R**3**. The third resistor **276** is also coupled to the second ground node **236**, GND_2. The second resistor **272** and the third resistor **276** have a resistance of about 1 kiloohm (kΩ).

[0032] The second input **264** of the difference amplifier **212** is coupled to a fourth resistor **280**, R**4** and to the second ground node **236**. The fourth resistor **280** is also coupled to an inverting input of the op-amp **268**, and to a fifth resistor **282**, R**5**. The fifth resistor **282** is also coupled to an output of the op-amp **268**, such that the fifth resistor **282** is a feedback resistor for the op-amp **268**. The fourth resistor **280** and the fifth resistor **282** have a resistance of about 1 kΩ. That is, the second resistor **272**, the third resistor **276**, the fourth resistor **280** and the fifth resistor **282** have the same resistance.

[0033] In operation, the high voltage supply **210** outputs a high voltage output, V_{OUT-HV} on the high side terminal **244** of the current sensor circuit **204**. The V_{OUT-HV} is a DC voltage. The high voltage output, V_{OUT-HV} is a voltage that is greater than about 400 V. In some examples, the V_{OUT-HV} is about 2000 V or more. This high voltage output, V_{OUT-HV} induces a DUT current, I_{DUT} that is provided to the DUT **208**. The difference amplifier **212** is configured to measure a voltage drop across the sensing resistor **252**, which corresponds to the DUT current, I_{DUT} and output a sensed voltage, V_{SNS} that varies as a function of a current across the sensing resistor **252** and corresponds to the DUT current, I_{DUT}. The isolation ADC **220** receives the sensed voltage, V_{SNS} and outputs a value encoded on a digital signal, ADC_{OUT} on the tester side that characterizes the voltage of the sensed voltage, V_{SNS}. The digital signal, ADC_{OUT} is provided to an external system.

[0034] Because the op-amp **268** is provided power by the isolation voltage, V_{ISO} output by the isolation DC-to-DC converter **216**, the op-amp **26** is floating with respect to the ground point of the DUT **208**, namely, the third ground node **240**. This indicates that the ground point of the op-amp **268**, namely the second ground node **236** is not directly connected to the ground reference (the third ground node **240**) of the DUT **208**. Instead, the op-amp **268** operates with an independent or isolated ground reference, the second ground node **236** which can be at a different electrical potential than the third ground node **240**. This configuration enables the op-amp **268** to function properly in environments with high common mode voltages, as it prevents the high voltage present at the DUT **208** from impacting operation of the op-amp **268**.

[0035] More particularly, as demonstrated by the current sensor circuit **100**, the sensing resistor **252** is coupled to the second node **256** of the current sensor circuit **204**, which is coupled to the high side terminal of the DUT **208** and to the second ground node **236**. Stated differently, a DUT side of the sensing resistor **252** is shorted to the second ground node **236**, GND_2.

[0036] The current sensor circuit **204** is configured such that with respect to the third ground node **240**, the voltage at the first node **248** of the current sensor circuit **204** $HV_{CURRENT_{SNS(+)}}$ is equal to V_{OUT-HV} and the voltage at the second node **256** of the current sensor circuit **204** $HV_{CURRENT_{SNS(-)}}$ is determined with Equation 1:

$$HV_{CURRENT_{SNS}(-)} = V_{OUT-HV} - I_{LEAK} * R_{I-SNS} \qquad \text{Equation 1}$$

[0037] Wherein:

[0038] R_{I-SNS} is the resistance of the sensing resistor (e.g., about 10 mΩ in some examples).

[0039] I_{LEAK} is the leakage current of the DUT that defines the current to the DUT during intervals of time that the DUT is an off-state.

[0040] Thus, with respect to the third ground node **240**, a difference voltage across the sensing resistor **252**, $V_{DIFF-GND3}$ is defined by Equation 2 and a common mode voltage, $V_{CM-GND3}$ is defined with Equation 3.

$$V_{DIFF-GND3} = I_{LEAK} * R_{I-SNS} \qquad \text{Equation 2}$$

$$V_{CM-GND3} = \frac{2 * VOUT_{HV} - Ileak * R_{I-SNS}}{2} \approx V_{OUT-HV} \qquad \text{Equation 3}$$

[0041] However, as noted, the difference amplifier **212** avoids the relatively high common mode voltage $V_{CM-GND3}$ by shorting the DUT side of the sensing resistor **252** (e.g., the second node **256**) with the high side terminal of the DUT **208**, creating two separate grounds, namely, the second ground node **236** and the third ground node **240**. Accordingly, with respect to the second ground node **236** and the op-amp **268**, the voltage difference across the sensing resistor **252**, $V_{DIFF-GND2}$ is defined with Equation 4 and the common mode voltage, $V_{CM-GND2}$ is defined with Equation 5.

$$V_{DIFF-GND2} = I_{LEAK} * R_{I-SNS} \qquad \text{Equation 4}$$

$$V_{CM-GND2} = (I_{LEAK} * R_{I-SNS})/2 \leq 5\text{ V} \qquad \text{Equation 5}$$

[0042] Thus, although the common mode voltage with respect to the third ground node **240** is approximately equal to the voltage at the high side terminal **244**, V_{OUT-HV}, the op-amp **268** can be measured with respect to the second ground node **236**, such that the common mode voltage of the op-amp **268**, $V_{CM-GND2}$ is less than about 5V. This avoids the need to have a relatively high common mode rejection error for components of the difference amplifier **212** (including the op-amp **268**). Accordingly, the current sensor circuit **204** provides an isolated current sensing solution that is safely useable with other low voltage circuits, such as other circuits on the tester side.

[0043] FIG. **3** illustrates an IC package level circuit **300** that is employable to implement the circuit **200** of FIG. **1**. The IC package level circuit **300** includes a high voltage supply **304** that is employable to implement the high voltage supply **210** of FIG. **2**. The IC package level circuit **300** includes a current sensor circuit **308**. The current sensor circuit **308** measures a DUT current, I_{DUT} provided to a DUT **310**.

[0044] The high voltage supply **304** is implemented on an IC package in some examples. Similarly, in some examples, the DUT **310** is implemented on an IC package. The DUT **310** has an on state, wherein the DUT current, I_{DUT} is at a peak value, and an off-state where the DUT current, I_{DUT} provided to the DUT **310** is a leakage current, I_{LEAK}.

[0045] The current sensor circuit **308** includes two IC packages, namely a first IC package **312** and a second IC package **316**. The first IC package **312** includes a difference amplifier **320** and an isolation ADC **324** (labeled ISO ADC). The difference amplifier **320** is employable to implement the difference amplifier **212** of FIG. **2** and/or the difference amplifier **108** of FIG. **1**. Thus, the difference amplifier **320** includes an op-amp (e.g., the op-amp **268** of FIG. **2**) configured in a differential configuration. The second IC package **316** includes an isolation DC-to-DC converter **328** (labeled ISO DC-DC) that is employable to implement the isolation ADC **220** of FIG. **2**. The current sensor circuit **308** also includes a sensing resistor **332**, R_{I-SNS} that is employable to implement the sensing resistor **252** of FIG. **2**, such that the sensing resistor **332** has a resistance of about 10 mΩ.

[0046] The IC package level circuit **300** includes a dashed line **336** that denotes a test side and a high voltage side of the IC package level circuit **300**, in the same manner as the dashed line **224** of FIG. **2**.

[0047] In some examples, the high voltage supply **304** is adapted to be coupled to a first input voltage, V_{IN1} and a first ground node **340**. In some examples, the first input voltage, V_{IN1} is a DC voltage of about 12 V or less. The isolation ADC **324** and the isolation DC-to-DC converter **328** are adapted to be coupled to a second input voltage, V_{IN2} and to the first ground node **340**. The second input voltage, V_{IN2} is a DC voltage of about 3 V to about 12 V (e.g., about 5V) in some examples.

[0048] Responsive to the second input voltage, V_{IN2}, the isolation DC-to-DC converter **328** outputs an isolation voltage, V_{ISO} that is about 5 V above a voltage of a second ground node **344**, GND_2. The isolation voltage, V_{ISO} is output to a high voltage terminal of the difference amplifier **320** and the isolation ADC **324**. A low voltage terminal of the difference amplifier **320** and the isolation ADC **324** are coupled to the second ground node **344**. In a similar manner, in response to the first input voltage, VIN_1, the high voltage supply **304** outputs a high voltage output, V_{OUT-HV} to a high side terminal **348** of the current sensor circuit **308**. The high voltage output, V_{OUT-HV} is a DC voltage of about 400 V to about 2000 V or more with respect to a third ground node **350** GND_3. Moreover, in some examples, the high voltage supply **304** is not coupled to the first ground node **340** and is coupled to the third ground node **350** such as in situations where the high voltage supply **304** is not an isolated voltage supply.

[0049] A low side terminal **352** of the current sensor circuit **308** is coupled to a high side terminal of the DUT **310**. The DUT **310** is also coupled to the third ground node **350**. The first ground node **340**, the second ground node **344** and the third ground node **350** are galvanically isolated from each other. Due to the high input impedance of the first input **356** and the second input **362** of the difference amplifier **320**, the sensing resistor **332** is in series with the DUT **310**.

[0050] The high side terminal **348** is also coupled to a first node **354** of the current sensor circuit **308**. The first node **354** of the sensor circuit is coupled to a first input **356** (e.g., a non-inverting input) of the difference amplifier **320** and to the sensing resistor **332**. The low side terminal **352** is coupled to a second node **358** of the current sensor circuit **308**. This second node **358** is also coupled to the second ground node **344**, the sensing resistor **332** and a second input **362** (e.g., an inverting input) of the difference amplifier **320**. The first input **356** of the difference amplifier **320** is employable to implement the first input **260** of the difference amplifier **212** of FIG. **2** and the second input **362** is employable to implement the second input **264** of the difference amplifier **212** of FIG. **2**.

[0051] The IC package level circuit **300** operates in the same manner as the circuit **200** of FIG. **2**. Thus, the high voltage supply **304** applies the output voltage, $V_{OUT\text{-}HV}$ to the high side terminal **348** of the current sensor circuit **308**, which induces the DUT current, I_{DUT} for the DUT **310**. The difference amplifier **320** measures a voltage drop across the sensing resistor **332** and outputs a sensing voltage, V_{SNS} that varies as a function of the current across the first ground node **232** and corresponds to the DUT current, I_{DUT} (and varies as a function of the voltage drop across the sensing resistor **332**). The sensing voltage, V_{SNS} is output to the isolation ADC **324**, which in turns outputs a digital value encoded in a digital signal, ADC_{OUT} that characterizes the sensed voltage, V_{SNS}. Thus, the digital value encoded in the ADC_{OUT} also characterizes the current to the DUT **310**.

[0052] As noted, the DUT **310** has an on-state, wherein the DUT current, I_{DUT} is presumed to be at a maximum. Also, the DUT **310** has an off-state where the DUT current, I_{DUT} provided to the DUT **310** is equal to the leakage current, I_{LEAK}. Moreover, as noted, the output voltage, $V_{OUT\text{-}HV}$ is a DC voltage of about 400 V to about 2000 V or more, including examples where the output voltage, $V_{OUT\text{-}HV}$ is about 1500 V. Thus, with respect to the third ground node **350**, the common mode voltage, $V_{CM\text{-}GND3}$ is about $V_{OUT\text{-}HV}$ as demonstrated in Equation 3. However, because the difference amplifier **320** employs the second ground node **344** as a reference and the difference amplifier **320** is powered by the isolation voltage, V_{ISO}, the common mode voltage for the difference amplifier **320** is measured with respect to the voltage at the second ground node **344**. Accordingly, as characterized in Equation 5, the common mode voltage with respect to the second ground node **344**, $V_{CM\text{-}GND2}$ is about 5 V or less. Thus, the difference amplifier **320** can be implemented with a standard op-amp arranged in a differential configuration, as illustrated in FIG. **2**.

[0053] FIG. **4** illustrates a chart **400** with measured results from implemented circuits, such as the circuit **200** of FIG. **2** and/or the IC package level circuit **300** of FIG. **3**. The chart **400** includes a ratio of a high voltage output (VHV) and a common mode voltage, VCM for a third ground node (GND**3**). This value could correspond, for example, to an output voltage of a high voltage supply, such as the high voltage supply **210** of FIG. **2** and/or the high voltage supply **304** of FIG. **3**. The chart **400** also plots a measured current, in microamperes (μA) to a DUT, such as the DUT **208** of FIG. **2** and/or the DUT **310** of FIG. **1**. The chart includes a measured current in μA for a current sensor circuit, such as the current sensor circuit **100** of FIG. **1**, the current sensor circuit **204** of FIG. **2** and/or the current sensor circuit **308** of FIG. **3**. The chart **400** further includes a current measure error in μA for the current sensor circuit. As demonstrated, the current measure error is less than 0.9 μA for each recorded output voltage. Thus, the chart **400** demonstrates that the current sensor circuit **100**, the current sensor circuit **204** of FIG. **2** and/or the current sensor circuit **308** of FIG. **3** are employable in situations where a high common mode voltage is present.

[0054] FIG. **5** is a flowchart of an example method **500** of sensing a current. In some examples, the method **500** is implemented with the current sensor circuit **100** of FIG. **1** and/or the current sensor circuit **204** of FIG. **2**. At block **510**, a difference amplifier (e.g., the difference amplifier **108** of FIG. **1**) senses a voltage drop across a sensing resistor (e.g., the sensing resistor **138** of FIG. **1**) having a first node (e.g., the first node **143** of FIG. **1**) adapted to be coupled to a high side terminal of a voltage supply (e.g., the high side terminal **142** of FIG. **1**) and a second node (e.g., the second node **144** of FIG. **1**) adapted to be coupled to a high side terminal of a DUT (e.g., the DUT **104** of FIG. **1**). A low side terminal of the DUT is coupled to a first ground node (e.g., the third ground node **154** of FIG. **1**), and the difference amplifier is coupled to the first node and the second node of the sensing resistor. A low side power terminal of the difference amplifier is coupled to a second ground node (e.g., the second ground node **134** of FIG. **1**), and the second ground node is coupled to the high side terminal of the DUT. The first and second ground nodes are galvanically isolated.

[0055] At block **515** the difference amplifier outputs a sensed voltage that varies as a function of a current across the sensing resistor in response to receiving an isolated voltage at a high side power terminal (e.g., the high side power terminal **126** of FIG. **1**) of the difference amplifier provided from an isolation DC (direct current)-to-DC converter (e.g., the isolation DC-to-DC converter **110** of FIG. **1**).

[0056] At block **520** an ADC (e.g., the isolation ADC **220** of FIG. **2**) receives the sensed voltage from the difference amplifier. At block **525** the isolation ADC outputs a digital value characterizing the current across the sensing resistor.

[0057] In this description, unless otherwise stated, "about" preceding a parameter means being within +/−10 percent of that parameter. Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

What is claimed is:

1. A current sensor circuit comprising:

a sensing resistor having a first node adapted to be coupled to a high side terminal of a voltage supply and a second node adapted to be coupled to a high side terminal of a DUT (device under test), wherein a low side terminal of the DUT is coupled to a first ground node;

a difference amplifier coupled to the first node and the second node of the sensing resistor, wherein a low side power terminal of the difference amplifier is coupled to a second ground node, and the second ground node is coupled to the high side terminal of the DUT; and

an isolation DC (direct current)-to-DC converter having an output coupled to a high side power terminal of the difference amplifier.

2. The current sensor circuit of claim **1**, wherein the first ground node and the second ground node are galvanically isolated.

3. The current sensor circuit of claim **1**, wherein the isolation DC-to-DC converter is coupled to a third ground node galvanically isolated from the first ground node and the second ground node.

4. The current sensor circuit of claim **3**, further comprising an isolation ADC (analog-to-digital converter) coupled to an output of the difference amplifier.

5. The current sensor circuit of claim **4**, wherein the isolation ADC and the difference amplifier are implemented on a first IC (integrated circuit) package and the isolation DC-to-DC converter is implemented on a second IC package.

6. The current sensor circuit of claim **5**, wherein the difference amplifier is configured to provide a sensed voltage at the output of the difference amplifier that varies as a function of a current across the sensing resistor.

7. The current sensor circuit of claim **6**, wherein the isolation ADC is configured to output a digital value characterizing the current across the sensing resistor.

8. The current sensor circuit of claim **4**, wherein the isolation ADC is coupled to the third ground node.

9. The current sensor circuit of claim **2**, wherein the difference amplifier is configured to output a sensed voltage that varies as a function of a current across the sensing resistor.

10. The current sensor circuit of claim **2**, wherein the voltage supply is configured to supply a voltage of at least **400** volts on the high side terminal of the voltage supply.

11. The current sensor circuit of claim **10**, wherein there is a voltage drop of about 5 volts or less between the first node and the second node.

12. The current sensor circuit of claim **2**, wherein the voltage supply is configured to supply a voltage of at least 1500 volts on the high side terminal of the voltage supply.

13. A current sensor circuit comprising:

a sensing resistor having a first node adapted to be coupled to a high side terminal of a voltage supply and a second node adapted to be coupled to a high side terminal of a DUT (device under test), wherein a low side terminal of the DUT is coupled to a first ground node;

a difference amplifier coupled to the first node and the second node of the sensing resistor, wherein a low side power terminal of the difference amplifier is coupled to a second ground node that is galvanically isolated from the first ground node, and the second ground node is coupled to the high side terminal of the DUT, the difference amplifier being configured to output a sensed voltage that varies as a function of a current across the sensing resistor; and

an isolation DC (direct current)-to-DC converter configured to provide an isolated voltage to a high side power terminal of the difference amplifier.

14. The current sensor circuit of claim **13**, wherein the first ground node and the second ground node have a potential difference.

15. The current sensor circuit of claim **14**, wherein the isolation DC-to-DC converter is coupled to a third ground node galvanically isolated from the first ground node and the second ground node.

16. The current sensor circuit of claim **14**, further comprising an isolation ADC (analog-to-digital converter) configured to receive the sensed voltage from the difference amplifier and to output a digital value characterizing the current across the sensing resistor.

17. The current sensor circuit of claim **14**, wherein the voltage supply is configured to supply a voltage of at least 400 volts on the high side terminal of the voltage supply, and there is a voltage drop of about 5 volts or less between the first node and the second node.

18. A method for sensing current, the method comprising:

sensing, by a difference amplifier, a voltage drop across a sensing resistor having a first node adapted to be coupled to a high side terminal of a voltage supply and a second node adapted to be coupled to a high side terminal of a DUT (device under test), wherein a low side terminal of the DUT is coupled to a first ground node, the difference amplifier is coupled to the first node and the second node of the sensing resistor, and a low side power terminal of the difference amplifier is coupled to a second ground node, and the second ground node is coupled to the high side terminal of the DUT; and

outputting, by the difference amplifier, a sensed voltage that varies as a function of a current across the sensing resistor in response to receiving an isolated voltage at a high side power terminal of the difference amplifier provided from an isolation DC (direct current)-to-DC converter.

19. The method of claim **18**, wherein the first ground node and the second ground node are galvanically isolated.

20. The method of claim **19**, further comprising:

receiving, at an isolation ADC (analog-to-digital converter) the sensed voltage from the difference amplifier; and

outputting, by the isolation ADC, a digital value characterizing the current across the sensing resistor.

* * * * *